

The LHS 1140 b catalog: what is certified and what is not

Two physics in this summary. Section 5 (the molecular $K_{zz} = 10^9$ family) was rewritten on 2026-09-26 from states certified under the atomic data as corrected on 2026-09-24 to 2026-09-26 (md/Update_EXHALE_stage3.md sections 17 to 27). Every other state, equivalent width, crossing and certificate below is of the physics before those corrections: none of those certificates holds under the corrected physics, and their equivalent widths are not current (the molecular family's crossing moved from He/H = 1.46 to 1.70 on the catalog grid). They are re-solved next, family by family.

Every model of the catalog is LHS 1140 b on 500 cells, solved by the partitioned stationary route with Well balanced: True, from the GJ 1132 spectrum placed at this planet's orbit. The binary of record of the rows solved up to 2026-09-22 is EXHALE_75d55d9d.x, md5 75d55d9d4fd0e748cd01d6e35713e34c; the rows added on 2026-09-23 and 2026-09-24 were solved and certified under EXHALE_69233669.x (the He/H = 9.7 rungs below 0.07 of the fiducial XUV) and EXHALE_46d5b9f4.x (the molecular $K_{zz} = 10^9$ rungs), as the table notes say. Five molecular $K_{zz} = 10^9$ rows were certified after the atomic-data corrections began: He/H = 1.45 on 2026-09-25 under EXHALE_b059a4be.x (the second of five rounds of corrections), and 1.52, 1.6, 1.8 and 2.13 on 2026-09-26 under the binary of record, EXHALE_7db5afe9.x, md5 7db5afe9b1d80ce757fcc07d1b499330 (section 5). The counts below are MEASURED by models/status.py inventory, re-run read-only on 2026-09-26: **157 state indexes, 100 certified** (the same on 2026-09-24; 143 and 92 on 2026-09-22). These counts are of the recorded certificates: all but the five molecular rows named above were measured under binaries before the atomic-data corrections, and none but the four of 2026-09-26 under the binary of record. A case counts as certified when its index names a latest_certified generation that no refused re-evaluation has marked stale. Section 3 and section 4 list the 103 models of the ladders and of diffusion_check; the remaining 54 indexes are 46 flux-closure iterates, two archival copies, one study directory and five run directories the low-XUV rungs left. Section 5, rewritten on 2026-09-26 under the corrected atomic physics: the molecular $K_{zz} = 10^9$ family straddles the observed equivalent width with two certified rungs (He/H = 1.6 and 1.8), the crossing moves with the grid to about 1.84 in the continuum limit, and the added Gaussian those solutions need to reproduce the width of the measured line is recorded.

Where each column comes from. du and $\|R\|$ are both read from the LAST (JFNK) done line of the case's run.log, so they are one pair printed by one solve and not two numbers from two places; where a case was last re-measured by an evaluation that took no step, that line is the solve preceding it, and where a case was never advanced there is none and the columns read --. The refusing rows of section 3 are read from certification.txt of the generation the case's index names latest_complete, and EVERY refusing entry of that certificate is listed, not only the first. $\log_{10} \dot{M}$ and the equivalent width are from models/status.py, which reads them from the case directory. The equivalent width is the red-pair integral over the vacuum window 10832.60 to 10834.20 Å taken on the observation's own wavelength samples: the instrument-convolved model is interpolated onto the 44 samples of the measurement inside the window and integrated with their trapezoid weights, the same operator that gives the observed 1.108 % Å (LHS1140b/he10830_equivalent_width.py, adopted 2026-09-23). Integrated on its own finer grid instead, a model differs by at most 1.5×10^{-4} % Å over the catalog, which moves 15 of the 91 certified rows below in the fourth decimal (MEASURED 2026-09-24). The XUV column is the factor the stellar spectrum is scaled by; K_{zz} is the eddy coefficient

of the base, *profile* where a photochemical column supplies $K_{zz}(p)$, and *none* where the base is well mixed. The *wind chemistry* column of every table below says what is solved ABOVE the base, atomic or molecular in the sense of section 1; what supplies the base itself is in the case name (scalar or photochem) and in the K_{zz} column.

1. Two axes: where the lower atmosphere comes from, and what the wind solves

A case name carries both axes, and they are independent. The first says what supplies the state at the base level, the second whether the wind above it carries a molecular network at all.

The lower boundary.

- **scalar**: the base is a few prescribed numbers at the level, read from `base.inp`: the pressure (1 microbar), the temperature (226 K), the base radius, the bound hydrogen fraction $q_{\text{H2,base}}$ and the element reservoirs. Nothing below the level is modeled.
- **scalarCNO**: the same scalar base with the C, N and O reservoirs of the photochemical column added at the matching level.
- **photochem**: the lower atmosphere is a Photochem column handed over as a PROFILE, the key Lower atmosphere profile: naming a file `lower_atmosphere_profile.dat`, which carries $T(p)$, $K_{zz}(p)$, the bound hydrogen fraction and the element reservoirs at the matching level $p_{\text{match}} = 1$ microbar. For the atomic groups the reservoir He/H is then solved by the elemental-flux closure (`src/utils/element_flux_closure.py`), which iterates the column and the wind against each other: the iterates are the `k<NN>` directories, and the composition is an output of that loop rather than an input.

The wind's chemistry.

- **atomic**: hydrogen and helium with their ionization stages and, where the group says so, trace metals. No molecular network is solved above the base, whatever the base carries.
- **molecular**: EXHALE's own network, Molecular chemistry: True with Molecular carrier transport: True, solving H_2 , H_2^+ , H_3^+ and HeH^+ in the wind with the H_2 carrier transported. The molecular base it starts from is the scalar handoff or the profile, by the other axis.

So `atomic_photochem...` is a Photochem column under an atomic wind, and `molecular_scalar...` is EXHALE's own molecular chemistry above a scalar base. The two are not alternatives for the same layer: the Photochem column is the atmosphere BELOW the matching level, and EXHALE's network is the chemistry ABOVE it. A `molecular_photochem` case uses both, the column below and the network above.

	wind solved atomic	wind solved with EXHALE's molecular network
scalar base (<code>base.inp</code>)	<code>atomic_scalar_*</code>	<code>molecular_scalar_*</code>
Photochem column (profile)	<code>atomic_photochem_*</code>	<code>molecular_photochem_*</code>

The mixing suffix is a third, smaller axis: wellmixed is no element diffusion, kzz1e9 is binary H/He diffusion at that eddy coefficient, and kzzprofile takes $K_{zz}(p)$ from the column.

2. By group

group	wind chemistry	He/H	XUV	K_{zz}	diff.	du reached	cases	cert.
diffusion_check	atomic	0.55	1	0.0 to 1.0e9	on	–	4	0
photochem_x0.01_kzzprofile	atomic	9.7	0.01	profile	on	9.4E-03	1	0
photochem_x0.10_kzzprofile	atomic	9.7	0.10	profile	on	5.0E-10	1	1
photochem_x0.15_kzzprofile	atomic	9.7	0.15	profile	on	4.7E-12	1	1
photochem_x0.20_kzzprofile	atomic	9.7	0.20	profile	on	2.0E-12	1	1
photochem_x0.25_kzzprofile	atomic	9.7	0.25	profile	on	1.9E-12	1	1
photochem_x0.30_kzzprofile	atomic	9.7	0.30	profile	on	1.3E-10	1	1
photochem_x0.33_kzzprofile	atomic	9.7	0.33	profile	on	1.6E-10	1	1
scalarCNO_kzz1e9	atomic	2.13	1	1.0e9	on	1.7E-11	1	1
scalar_gj699_wellmixed	atomic	0.042 to 1000	1	none	off	1.8E-13 to 1.6E-11	6	6
scalar_kzz0	atomic	0.55 to 3.9	1	0.0	on	2.4E-11 to 1.2E-10	6	6
scalar_kzz1e10	atomic	0.55 to 1.29	1	1.0e10	on	2.0E-11 to 2.7E-11	4	4
scalar_kzz1e11	atomic	0.55 to 0.93	1	1.0e11	on	1.8E-11 to 1.9E-10	4	4
scalar_kzz1e5	atomic	2.6 to 3.93	1	1.0e5	on	2.5E-11 to 4.8E-11	5	5
scalar_kzz1e6	atomic	0.55 to 3.74	1	1.0e6	on	1.9E-11 to 6.8E-11	6	6
scalar_kzz1e7	atomic	0.55 to 3.18	1	1.0e7	on	1.6E-12 to 1.3E-10	4	4
scalar_kzz1e8	atomic	0.55 to 2.41	1	1.0e8	on	5.5E-11 to 7.9E-11	4	4
scalar_kzz1e9	atomic	0.55 to 9.7	1	1.0e9	on	9.3E-13 to 6.1E-11	7	7
scalar_wellmixed	atomic	0.083 to 1000	1	none	off	1.4E-12 to 1.2E-10	9	9
scalar_x0.015_kzz1e9	atomic	2.13	0.015	1.0e9	on	1.8E-10	1	1
scalar_x0.01_kzz1e9	atomic	2.13, 9.7	0.01	1.0e9	on	4.4E-12 to 2.2E-10	2	1
scalar_x0.02_kzz1e9	atomic	2.13	0.02	1.0e9	on	8.3E-11	1	1
scalar_x0.03_kzz1e9	atomic	2.13	0.03	1.0e9	on	4.4E-11	1	1
scalar_x0.05_kzz1e9	atomic	2.13	0.05	1.0e9	on	6.4E-11	1	1
scalar_x0.06_kzz1e9	atomic	9.7	0.06	1.0e9	on	2.8E-12	1	0
scalar_x0.0624_kzz1e9	atomic	9.7	0.0624	1.0e9	on	3.1E-12	1	0
scalar_x0.065_kzz1e9	atomic	9.7	0.065	1.0e9	on	2.6E-12	1	1
scalar_x0.0662_kzz1e9	atomic	9.7	0.0662	1.0e9	on	2.5E-12	1	1
scalar_x0.0675_kzz1e9	atomic	9.7	0.0675	1.0e9	on	3.3E-12	1	1
scalar_x0.07_kzz1e9	atomic	2.13, 9.7	0.07	1.0e9	on	7.0E-12 to 5.1E-11	2	2
scalar_x0.10_kzz1e9	atomic	2.13, 9.7	0.10	1.0e9	on	4.4E-12 to 1.5E-11	2	2
scalar_x0.15_kzz1e9	atomic	2.13, 9.7	0.15	1.0e9	on	8.2E-12 to 9.1E-12	2	2
scalar_x0.20_kzz1e9	atomic	2.13, 9.7	0.20	1.0e9	on	5.5E-12 to 1.1E-11	2	2
scalar_x0.25_kzz1e9	atomic	2.13, 9.7	0.25	1.0e9	on	5.2E-12 to 1.7E-11	2	2
scalar_x0.30_kzz1e9	atomic	2.13, 9.7	0.30	1.0e9	on	3.2E-12 to 5.6E-12	2	2
scalar_x0.33_kzz1e9	atomic	2.13, 9.7	0.33	1.0e9	on	2.8E-12 to 3.4E-12	2	2
photochem_kzzprofile	molecular	9	1	profile	on	1.4E-02	1	0
scalar_kzz1e9	molecular	0.083 to 9.7	1	1.0e9	on	1.4E-12 to 4.5E-11	8	7
scalar_wellmixed	molecular	0.083, 0.55	1	none	off	9.1E-13 to 6.6E-12	2	0
total							103	91

Besides these, the catalog holds 46 flux-closure iterates k<NN>, two archival copies, one study directory and five run directories of the low-XUV rungs (each run keeps the case files it ended with, and its index counts in the inventory). Nine of the iterates carry a certificate; the other 37 were written by runs that asked for no time integration, so they are states that were post-processed and never advanced, and they are not failures of a solve.

Two molecular cases carry no index at all and are not counted in the table: molecular_scalar_gj1132_wellmixed/HeH 2.13 and molecular_photochem_gj1132_kzzprofile/HeH2.09 were attempted and published no state (section 3).

3. The models that are not certified

Twelve of the 103 models carry no certificate. Each was solved, or attempted, or written without being advanced at all, and the certificate of the generation its index names refuses it. The last column lists every entry of that certificate that stands above its tolerance, with the measure and the cell, in the certificate's own order.

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\ R\ $	the rows the certificate refuses on
diffusion_check/scalar_kzz0_HeH0.55	atomic	0.55	1	0.0	on	–	–	mass 1.541 at cell 1; momentum 7.593e-3 at cell 500; energy 1.583 at cell 5; elemental He/H 2.546e-4 at cell 500

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\ R\ $	the rows the certificate refuses on
diffusion_check/scalar_kzz1e10_HeH0.55	atomic	0.55	1	1.0e10	on	–	–	mass 1.541 at cell 1; momentum 2.466e-3 at cell 500; energy 1.907 at cell 5; elemental He/H 9.644e-5 at cell 500
diffusion_check/scalar_kzz1e8_HeH0.55	atomic	0.55	1	1.0e8	on	–	–	mass 1.542 at cell 1; momentum 5.150e-3 at cell 500; energy 1.952 at cell 5; elemental He/H 1.790e-4 at cell 500
diffusion_check/scalar_kzz1e9_HeH0.55	atomic	0.55	1	1.0e9	on	–	–	mass 1.542 at cell 1; momentum 3.719e-3 at cell 500; energy 1.917 at cell 5; elemental He/H 1.354e-4 at cell 500
photochem_x0.01_kzzprofile/HeH9.7	atomic	9.7	0.01	profile	on	9.356E-03	1.163E+00	mass 5.880e-1 at cell 11; momentum 7.774e-7 at cell 495; energy 1.949 at cell 7; elemental He/H 3.538e-2 at cell 389
scalar_x0.01_kzz1e9/HeH9.7	atomic	9.7	0.01	1.0e9	on	4.424E-12	9.978E-01	mass 9.806e-1 at cell 2; momentum 8.724e-6 at cell 1; energy 1.071 at cell 1
scalar_x0.06_kzz1e9/HeH9.7	atomic	9.7	0.06	1.0e9	on	2.785E-12	5.589E-07	elemental He/H 2.508e-2 at cell 333
scalar_x0.0624_kzz1e9/HeH9.7	atomic	9.7	0.0624	1.0e9	on	3.086E-12	8.091E-08	elemental He/H 1.435e-2 at cell 333
photochem_kzzprofile/HeH9	molecular	9	1	profile	on	1.406E-02	3.167E-01	mass 1.041 at cell 2; momentum 1.450e-3 at cell 1; energy 1.021 at cell 1; H ₂ carrier 1.000 at cell 432; elemental He/H 1.494e-4 at cell 217
scalar_kzz1e9/HeH0.083	molecular	0.083	1	1.0e9	on	4.879E-12	1.369E-08	H ₂ carrier 8.150e-4 at cell 499; elemental He/H 2.503e-4 at cell 280
scalar_wellmixed/HeH0.083	molecular	0.083	1	none	off	6.592E-12	2.701E-08	H ₂ carrier 7.371e-2 at cell 500
scalar_wellmixed/HeH0.55	molecular	0.55	1	none	off	9.086E-13	1.062E-08	energy 1.000 at cell 246; H ₂ carrier 2.363e-2 at cell 306

Read by mechanism, the twelve fall into four kinds.

- **The hydrodynamic Newton makes no progress** on the two models that remain at 0.01 of the fiducial XUV. The composition relaxes while the hydrodynamic rows stand still between passes. These are the models of the open solver item. The third model of that group, scalar_x0.01_kzz1e9/HeH2.13, left this table on 2026-09-21: an XUV continuation down the ladder 0.07, 0.05, 0.03, 0.02, 0.015, 0.01 reached a certified state of it, 17 to 19 passes a rung, every rung ending $\text{info} = 0$. It was reached by continuation and not by a better solver, and nothing establishes either way whether the two models still here have a stationary solution at all.
- **The elemental He/H row refuses** the two He/H = 9.7 rungs at 0.06 and 0.0624 of the fiducial XUV, at cell 333, with the hydrodynamic rows inside their tolerances: continued from the certified 0.065 rung, each was refused by the progress control within five passes, the row rising (2026-09-23).
- **The composition half of the solve refuses** the four molecular models: the H₂ carrier balance in all four, and in photochem_kzzprofile/HeH9 also the base cell's hydrodynamic rows. The wind itself is converged in three of them. The next subsection says what was measured.
- **Nothing was solved** in the four diffusion_check models: they were post-processed without a time integration, so they have no du and no $\|R\|$. Their certificates were measured all the same, and refuse on the hydrodynamic rows and the elemental row, which is what an unadvanced state is expected to do; they are not failures of a solve.

What the two numbers say. du is the fractional spread of the mass flux $\rho v r^2$ over the wind window, the quantity a marching run stops on, and $\|R\|$ is the stationary operator's residual norm. The recipe of this code, and the practice of the literature it follows, calls a wind converged at $du < 10^{-3}$. MEASURED on the models that carry no certificate, naming the wind chemistry because the same short name occurs in both families: the molecular scalar_kzz1e9/HeH0.083 reaches $du = 4.9 \times 10^{-12}$ with $\|R\| = 1.4 \times 10^{-8}$, the molecular scalar_wellmixed/HeH0.083 6.6×10^{-12} and 2.7×10^{-8} , the molecular scalar_wellmixed/HeH0.55 9.1×10^{-13} and 1.1×10^{-8} , and the atomic scalar_x0.01_kzz1e9/HeH9.7 4.4×10^{-12} with $\|R\| = 1.0$, and the atomic He/H = 9.7 rungs at 0.06 and 0.0624 2.8×10^{-12} and 3.1×10^{-12} with 5.6×10^{-7} and 8.1×10^{-8} . By the flux-spread criterion these are converged winds, and three of them also sit at a residual norm of 10^{-8} .

What refuses them is this code's own row-by-row certification, which is stricter: it measures every equation of the inventory cell by cell and names each one that stands above its tolerance. Over the twelve the elemental He/H row appears in nine of them, the hydrodynamic energy row in eight, the mass and momentum rows in seven each and the H₂ carrier balance in the wind in four (MEASURED on the table above). A state that meets the flux-spread criterion and fails one such row is a meaningful solution by the standard other work uses, and the table says which rows keep it from this one.

Why the molecular winds do not certify

The verdict. The failure is of the method, not a finding that these winds have no stationary state. The partitioned route alternates two solves in every outer pass: the hydrodynamic Newton solve at fixed composition, then the relaxation of the composition (the element rows and the H_2 carrier) at fixed wind. For the six molecular cases below the alternation does not approach a joint fixed point. The one route that solves both halves together fails in its linear solve. Every measurement of this subsection was made under binaries before the atomic-data corrections, among them EXHALE_3146d11b.x (2026-09-18) and the tree's binary of 2026-09-21 (md5 89ec6714...). Whether the same cases behave the same way under EXHALE_7db5afe9.x was not measured.

The six cases. The four of the table above, and the two with no index (READ from each case's not_solved.md and from md/lhs1140b_stationary_L34c_20260918.md). The two with no index were started from the certified atomic wind of the same composition. Each was stopped at the six-hour ceiling:

- molecular_scalar_gj1132_wellmixed/HeH2.13: the carrier H_2 row flattened at 4.3×10^{-1} after seven outer passes.
- molecular_photochem_gj1132_kzzprofile/HeH2.09: the row banded between 0.19 and 0.44 over twelve outer passes without a trend.

What six outer passes showed, measured on 2026-09-21. Each of four cases was advanced six outer passes from its stored state (md/lhs1140b_alternation_audit_20260921.md). Three did not approach a fixed point, by two different mechanisms, and one did:

- **The composition update takes the wind out of solution** (wellmixed/HeH0.083 and wellmixed/HeH0.55).
 - One bounded composition update moved the mass flux by 0.985 of its own column scale.
 - From pass 2 on, the hydrodynamic solve returned `info = 2`, with its mass and energy rows seven decades above tolerance, and never recovered.
 - In wellmixed/HeH0.55 the H_2 front also walked outward: the refusing cell moved from 306 to 349 ($r = 1.95$ to $3.02 R_p$) while the row rose by a factor 4.5.
- **A drift under a saturated movement bound** (kzz1e9/HeH0.083).
 - The carrier relaxation ended on its movement bound in every pass, at cells 418 to 426 ($r = 8.0$ to $8.7 R_p$).
 - The gated carrier row rose from 8.2×10^{-4} to 2.9×10^{-3} , while the hydrodynamic rows stayed within tolerance.
 - Relocating the bound does not help: the bound then acts at the edge of the mask, and the refusing row rises in every such run (2026-09-17). So the bound is not the defect.
- **Approaching, and stopped by the budget** (kzz1e9/HeH0.55): the joint residual fell 3.7 decades in five passes. This case was then finished and certified on 2026-09-24 (section 4).

The route that solves both halves together fails. Two measurements show it, on 2026-09-17 and 2026-09-21 (md/lhs1140b_coupled_trial_20260921.md). With Coupled carrier solve: True, the element and H_2 carrier rows become unknowns of the Newton solve.

- On kzz1e9/HeH0.083, that solve spent 41 trust-region iterations, 6632 residual evaluations and 7172 s. It moved the merit only from 9.720×10^{-2} to 9.718×10^{-2} , stagnated (`info = 2`) and handed back the state it started from.
- On a case the alternation certifies, all 13 linear solves exhausted their 40-vector Krylov subspace at a relative residual of 0.26 against the 0.1 asked.

The missing piece is a preconditioner that holds the species rows. It is the open item that owns these cases.

Explanations excluded by measurement. These are, in order:

- *An outer boundary that traps H_2 .* At cell 500 the diffusive face is exactly zero, because the carrier's ghost repeats the last value. Opening that face makes the refusing row worse, by factors 1.12 to 182: H_2 is heavier than the mean particle of the ionized wind, so the settling drift the open face admits points inward.
- *An inexact inner solve.* In each pass the element relaxation reached the fixed point of its own operator.
- *A two-state cycle.* The two-pass state difference is never small against the one-pass differences it spans.
- *A refusing cell that only appears to move.* It is fixed at cell 500 from pass 2 on in three of the four cases.

What does certify. The same network, with binary H/He diffusion at $K_{zz} = 10^9$, certifies from He/H = 0.55 to 9.7 (sections 4 and 5). What separates those states from the refused ones, beyond the measurements above, was not established.

4. The models that are certified

Ninety-one models carry a certificate. The table lists each with the mass-loss rate and the He I 10830 equivalent width the catalog reports for it. Eight of these rows are new since 2026-09-22: the He/H = 9.7 rungs at 0.0675, 0.0662 and 0.065 of the fiducial XUV (certified on 2026-09-23 under EXHALE_69233669.x, continued from the 0.07 rung) and the molecular rows at He/H = 0.55, 1.45, 1.52, 1.6 and 1.8 (certified on 2026-09-24 under EXHALE_46d5b9f4.x; section 5); of those five, 1.45 was certified again on 2026-09-25 and 1.52, 1.6 and 1.8 on 2026-09-26, and the table gives those later states (notes a and b). Seven rows were new between 2026-09-20 and 2026-09-22: the six rungs of the 2026-09-21 XUV continuation on the He/H = 2.13 branch at 0.015, 0.02, 0.03, 0.05 and 0.07 of the fiducial spectrum together with the 0.07 rung on the He/H = 9.7 branch, and the 0.01 rung on the He/H = 2.13 branch that the continuation certified. The molecular rows at He/H = 2.13 and 9.7 certified on 2026-09-20, when every active equation of the inventory was measured on their states and found within tolerance; 2.13 was certified again on 2026-09-26 (note a).

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\ R\ $	$\log_{10} \dot{M}$	EW
photochem_x0.10_kzzprofile/HeH9.7	atomic	9.7	0.10	profile	on	4.973E-10	1.022E-07	6.840	0.2674
photochem_x0.15_kzzprofile/HeH9.7	atomic	9.7	0.15	profile	on	4.692E-12	6.210E-08	7.020	0.4822
photochem_x0.20_kzzprofile/HeH9.7	atomic	9.7	0.20	profile	on	2.002E-12	6.628E-08	7.160	0.6711
photochem_x0.25_kzzprofile/HeH9.7	atomic	9.7	0.25	profile	on	1.900E-12	6.245E-08	7.260	0.8428
photochem_x0.30_kzzprofile/HeH9.7	atomic	9.7	0.30	profile	on	1.261E-10	3.190E-08	7.350	1.0097
photochem_x0.33_kzzprofile/HeH9.7	atomic	9.7	0.33	profile	on	1.644E-10	3.457E-08	7.390	1.0979
scalarCNO_kzz1e9/HeH2.13	atomic	2.13	1	1.0e9	on	1.739E-11	1.352E-08	8.870	1.3721
scalar_gj699_wellmixed/HeH0.042	atomic	0.042	1	none	off	2.983E-13	1.341E-07	8.570	0.9091
scalar_gj699_wellmixed/HeH0.046	atomic	0.046	1	none	off	1.827E-13	9.505E-08	8.570	0.9857
scalar_gj699_wellmixed/HeH0.050	atomic	0.050	1	none	off	1.587E-11	1.414E-07	8.570	1.0609
scalar_gj699_wellmixed/HeH0.083	atomic	0.083	1	none	off	7.470E-13	1.320E-07	8.570	1.6271
scalar_gj699_wellmixed/HeH1	atomic	1	1	none	off	5.765E-13	8.372E-08	8.550	4.5248
scalar_gj699_wellmixed/HeH1000	atomic	1000	1	none	off	8.965E-13	7.482E-09	8.420	2.8724
scalar_kzz0/HeH0.55	atomic	0.55	1	0.0	on	1.234E-10	1.707E-08	7.840	0.0305
scalar_kzz0/HeH2.6	atomic	2.6	1	0.0	on	4.029E-11	1.344E-08	7.860	0.9549
scalar_kzz0/HeH3.0	atomic	3.0	1	0.0	on	3.264E-11	2.215E-08	7.870	1.0798
scalar_kzz0/HeH3.5	atomic	3.5	1	0.0	on	2.429E-11	1.458E-08	7.870	1.2175
scalar_kzz0/HeH3.7	atomic	3.7	1	0.0	on	4.900E-11	1.776E-08	7.870	1.2698
scalar_kzz0/HeH3.9	atomic	3.9	1	0.0	on	4.557E-11	2.821E-08	7.870	1.3168
scalar_kzz1e10/HeH0.55	atomic	0.55	1	1.0e10	on	1.963E-11	1.870E-08	7.860	0.6463
scalar_kzz1e10/HeH1.06	atomic	1.06	1	1.0e10	on	1.994E-11	2.280E-08	7.860	1.0456
scalar_kzz1e10/HeH1.15	atomic	1.15	1	1.0e10	on	2.240E-11	1.463E-08	7.860	1.1031
scalar_kzz1e10/HeH1.29	atomic	1.29	1	1.0e10	on	2.711E-11	1.336E-08	7.860	1.1867
scalar_kzz1e11/HeH0.55	atomic	0.55	1	1.0e11	on	1.769E-11	2.546E-08	7.860	0.8065
scalar_kzz1e11/HeH0.795	atomic	0.795	1	1.0e11	on	5.628E-11	1.651E-08	7.860	1.0302
scalar_kzz1e11/HeH0.865	atomic	0.865	1	1.0e11	on	7.733E-11	1.951E-08	7.860	1.0866
scalar_kzz1e11/HeH0.93	atomic	0.93	1	1.0e11	on	1.864E-10	2.097E-08	7.860	1.1364
scalar_kzz1e5/HeH2.6	atomic	2.6	1	1.0e5	on	3.749E-11	1.394E-08	7.870	0.9631
scalar_kzz1e5/HeH3.0	atomic	3.0	1	1.0e5	on	2.952E-11	1.531E-08	7.870	1.0880
scalar_kzz1e5/HeH3.35	atomic	3.35	1	1.0e5	on	2.488E-11	1.509E-08	7.870	1.1862
scalar_kzz1e5/HeH3.64	atomic	3.64	1	1.0e5	on	4.781E-11	1.764E-08	7.870	1.2627
scalar_kzz1e5/HeH3.93	atomic	3.93	1	1.0e5	on	4.160E-11	1.552E-08	7.870	1.3314
scalar_kzz1e6/HeH0.55	atomic	0.55	1	1.0e6	on	1.895E-11	2.047E-08	7.840	0.0590
scalar_kzz1e6/HeH2.4	atomic	2.4	1	1.0e6	on	2.785E-11	1.570E-08	7.860	0.9465
scalar_kzz1e6/HeH2.8	atomic	2.8	1	1.0e6	on	4.856E-11	1.407E-08	7.870	1.0804
scalar_kzz1e6/HeH3.19	atomic	3.19	1	1.0e6	on	4.002E-11	2.368E-08	7.870	1.1948
scalar_kzz1e6/HeH3.46	atomic	3.46	1	1.0e6	on	3.570E-11	1.742E-08	7.870	1.2669
scalar_kzz1e6/HeH3.74	atomic	3.74	1	1.0e6	on	6.781E-11	2.390E-08	7.870	1.3388
scalar_kzz1e7/HeH0.55	atomic	0.55	1	1.0e7	on	1.289E-10	2.116E-08	7.850	0.1586

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\ R\ $	$\log_{10} \dot{M}$	EW
scalar_kzz1e7/HeH2.70	atomic	2.70	1	1.0e7	on	2.053E-12	1.497E-08	7.870	1.1996
scalar_kzz1e7/HeH2.94	atomic	2.94	1	1.0e7	on	1.639E-12	1.420E-08	7.870	1.2724
scalar_kzz1e7/HeH3.18	atomic	3.18	1	1.0e7	on	4.895E-11	3.913E-08	7.870	1.3265
scalar_kzz1e8/HeH0.55	atomic	0.55	1	1.0e8	on	5.543E-11	2.155E-08	7.850	0.3089
scalar_kzz1e8/HeH2.05	atomic	2.05	1	1.0e8	on	7.949E-11	1.428E-08	7.870	1.1513
scalar_kzz1e8/HeH2.23	atomic	2.23	1	1.0e8	on	7.229E-11	1.895E-08	7.870	1.2207
scalar_kzz1e8/HeH2.41	atomic	2.41	1	1.0e8	on	5.482E-11	1.624E-08	7.870	1.2826
scalar_kzz1e9/HeH0.55	atomic	0.55	1	1.0e9	on	6.072E-11	1.518E-08	7.850	0.4794
scalar_kzz1e9/HeH1.50	atomic	1.50	1	1.0e9	on	2.944E-11	1.646E-08	7.870	1.0996
scalar_kzz1e9/HeH1.60	atomic	1.60	1	1.0e9	on	4.581E-11	1.313E-08	7.870	1.1511
scalar_kzz1e9/HeH1.70	atomic	1.70	1	1.0e9	on	3.779E-11	2.343E-08	7.870	1.1969
scalar_kzz1e9/HeH2.13	atomic	2.13	1	1.0e9	on	2.462E-11	1.700E-08	7.870	1.3749
scalar_kzz1e9/HeH4.0	atomic	4.0	1	1.0e9	on	9.341E-13	1.470E-08	7.870	1.8229
scalar_kzz1e9/HeH9.7	atomic	9.7	1	1.0e9	on	5.192E-11	2.009E-08	7.860	2.0592
scalar_wellmixed/HeH0.083	atomic	0.083	1	none	off	9.755E-12	2.543E-08	7.840	0.3798
scalar_wellmixed/HeH0.40	atomic	0.40	1	none	off	1.442E-12	2.641E-08	7.850	1.1011
scalar_wellmixed/HeH0.42	atomic	0.42	1	none	off	3.329E-12	1.823E-08	7.850	1.1324
scalar_wellmixed/HeH0.44	atomic	0.44	1	none	off	2.623E-12	1.723E-08	7.850	1.1626
scalar_wellmixed/HeH0.55	atomic	0.55	1	none	off	1.618E-12	1.929E-08	7.850	1.3118
scalar_wellmixed/HeH1	atomic	1	1	none	off	4.004E-12	1.485E-08	7.860	1.6958
scalar_wellmixed/HeH10	atomic	10	1	none	off	1.205E-10	1.281E-08	7.850	1.9347
scalar_wellmixed/HeH100	atomic	100	1	none	off	1.684E-12	1.075E-08	7.860	1.9644
scalar_wellmixed/HeH1000	atomic	1000	1	none	off	1.744E-12	1.567E-08	7.850	1.9197
scalar_x0.015_kzz1e9/HeH2.13	atomic	2.13	0.015	1.0e9	on	1.775E-10	9.233E-07	5.970	0.0000
scalar_x0.01_kzz1e9/HeH2.13	atomic	2.13	0.01	1.0e9	on	2.220E-10	3.154E-07	5.790	0.0000
scalar_x0.02_kzz1e9/HeH2.13	atomic	2.13	0.02	1.0e9	on	8.304E-11	1.096E-07	6.100	0.0000
scalar_x0.03_kzz1e9/HeH2.13	atomic	2.13	0.03	1.0e9	on	4.353E-11	1.169E-07	6.280	0.0000
scalar_x0.05_kzz1e9/HeH2.13	atomic	2.13	0.05	1.0e9	on	6.354E-11	2.969E-07	6.510	0.0000
scalar_x0.065_kzz1e9/HeH9.7	atomic	9.7	0.065	1.0e9	on	2.649E-12	5.760E-08	6.610	0.0132
scalar_x0.0662_kzz1e9/HeH9.7	atomic	9.7	0.0662	1.0e9	on	2.534E-12	2.452E-08	6.620	0.0172
scalar_x0.0675_kzz1e9/HeH9.7	atomic	9.7	0.0675	1.0e9	on	3.317E-12	3.863E-08	6.630	0.0239
scalar_x0.07_kzz1e9/HeH2.13	atomic	2.13	0.07	1.0e9	on	5.100E-11	1.976E-07	6.660	0.0000
scalar_x0.07_kzz1e9/HeH9.7	atomic	9.7	0.07	1.0e9	on	6.972E-12	9.931E-08	6.640	0.0445
scalar_x0.10_kzz1e9/HeH2.13	atomic	2.13	0.10	1.0e9	on	1.464E-11	3.886E-07	6.820	0.0001
scalar_x0.10_kzz1e9/HeH9.7	atomic	9.7	0.10	1.0e9	on	4.424E-12	4.362E-07	6.810	0.2353
scalar_x0.15_kzz1e9/HeH2.13	atomic	2.13	0.15	1.0e9	on	9.116E-12	1.239E-07	7.000	0.0003
scalar_x0.15_kzz1e9/HeH9.7	atomic	9.7	0.15	1.0e9	on	8.205E-12	1.162E-07	7.000	0.4514
scalar_x0.20_kzz1e9/HeH2.13	atomic	2.13	0.20	1.0e9	on	1.072E-11	1.165E-07	7.130	0.0022
scalar_x0.20_kzz1e9/HeH9.7	atomic	9.7	0.20	1.0e9	on	5.463E-12	8.109E-08	7.130	0.6294
scalar_x0.25_kzz1e9/HeH2.13	atomic	2.13	0.25	1.0e9	on	5.191E-12	1.806E-07	7.240	0.0700
scalar_x0.25_kzz1e9/HeH9.7	atomic	9.7	0.25	1.0e9	on	1.677E-11	1.044E-07	7.230	0.7913
scalar_x0.30_kzz1e9/HeH2.13	atomic	2.13	0.30	1.0e9	on	3.218E-12	6.850E-08	7.320	0.1894
scalar_x0.30_kzz1e9/HeH9.7	atomic	9.7	0.30	1.0e9	on	5.587E-12	6.702E-08	7.310	0.9395
scalar_x0.33_kzz1e9/HeH2.13	atomic	2.13	0.33	1.0e9	on	2.761E-12	4.566E-08	7.360	0.2555
scalar_x0.33_kzz1e9/HeH9.7	atomic	9.7	0.33	1.0e9	on	3.376E-12	9.900E-08	7.360	1.0231
scalar_kzz1e9/HeH0.55	molecular	0.55	1	1.0e9	on	2.822E-12	9.541E-09	7.590	0.1015
scalar_kzz1e9/HeH1.45 ^b	molecular	1.45	1	1.0e9	on	4.524E-11	7.974E-09	7.900	0.9387
scalar_kzz1e9/HeH1.52 ^a	molecular	1.52	1	1.0e9	on	2.068E-12	7.337E-09	7.890	1.0149
scalar_kzz1e9/HeH1.6 ^a	molecular	1.6	1	1.0e9	on	1.441E-12	1.167E-08	7.900	1.0577
scalar_kzz1e9/HeH1.8 ^a	molecular	1.8	1	1.0e9	on	4.134E-12	8.818E-09	7.910	1.1550
scalar_kzz1e9/HeH2.13 ^a	molecular	2.13	1	1.0e9	on	3.364E-11	1.266E-08	7.930	2.2902
scalar_kzz1e9/HeH9.7 ^c	molecular	9.7	1	1.0e9	on	3.491E-11	2.580E-08	7.940	2.4060

^a Certified on 2026-09-26 under the binary of record, EXHALE_7db5afe9.x: the only rows of this table under the corrected atomic physics (section 5). ^b Certified on 2026-09-25 under EXHALE_b059a4be.x, after the second of the five rounds of corrections: neither the physics before them nor the corrected one. ^c The certified generation is the 2026-09-20 evaluation of the state written on 2026-09-19. A re-solve made later on 2026-09-20 did not certify (the H₂ carrier row 5.1×10^{-4} at cell 262 after 40 passes); its not_solved.md stays beside the case and describes that re-solve, not the state listed here. The du and $\|R\|$ of this row are that re-solve's, as the rule of the column paragraph above gives them. Every unmarked row is of the physics before the corrections.

5. The molecular $K_{zz} = 10^9$ family across the observed line

What changed, and on which binary. The atomic data of EXHALE were checked against their original sources on 2026-09-24 to 2026-09-26 (md/Update_EXHALE_stage3.md sections 17 to 26). The corrections that move this line most are the electron-impact de-excitation $2^3S \rightarrow 1^1S$ (the detailed-balance reverse of the excitation, omitted by the metastable network before), the He I collision strengths of Bray et al. (2000) in place of an extrapolated fit, the case-A/B escape of the H ground captures, and the He⁺ + H charge exchange, whose non-radiative channel the Kingdon & Ferland (1996) fit had overstated by a factor of about 200 at 10^4 K. Every number of this section is from states certified under that physics (EXHALE_59c6be36.x; the binary of record of the catalog rows, EXHALE_7db5afe9.x, differs from it only in refusing, rather than stopping at, an unclosed lower ghost of a trial state, and gives byte-identical

results on every run that does not meet it). The values of this section before the audit are in sections 11 to 15 of the update log. The He/H = 0.55 and 9.7 molecular rows of the catalog were not re-solved and are of the physics before the audit; the 1.45 row is of the second round of corrections (EXHALE_b059a4be.x, EW 0.9387). Under the third round (EXHALE_50aa18d7.x) neither its re-solve nor its rung from 1.52 certified before its time ceiling, and it was not attempted under the binary of record.

The rungs. He/H = 1.8 and 2.13 were re-solved from their certified states of the previous physics, and 1.6 and 1.52 as rungs from 1.8 (the rung to 1.45, made under the third round, did not certify: the H₂ carrier row held at 10⁻¹ through four passes). Each certified state was published into the catalog through run_case.sh, and every re-measurement reproduced its claim. MEASURED:

He/H	passes	EW [% Å]	log ₁₀ $\dot{M}$ [g s ⁻¹]
1.52	24	1.0149	7.89
1.6	25	1.0577	7.90
1.8	23	1.1550	7.91
2.13	21	1.2902	7.93
observed		1.108 ± 0.030	

The certified rungs 1.6 and 1.8 straddle the observed EW, and the log-log chord crosses 1.108 at He/H = 1.703 on the catalog grid (DERIVED); before the audit the crossing was 1.464. The profiles move monotonically with He/H (Fig. 3).

The grid of the crossing. The three straddling compositions were solved again on 1000 and 2000 cells (the base block and every other spacing halved at each step, the outer radius unchanged); all nine states certify, and every evaluation. MEASURED, with the continuum DERIVED by Richardson extrapolation at the measured order:

	500 cells	1000 cells	2000 cells	continuum
EW, He/H = 1.6	1.0577	1.0297	1.0150	0.999
EW, He/H = 1.8	1.1550	1.1249	1.1094	1.093
EW, He/H = 2.13	1.2902	1.2571	1.2403	1.223
crossing of 1.108, EW	1.703	1.764	1.797	1.838
crossing of 1.108, broadened EW	1.781	1.851	1.890	1.933

The EW converges at first order in the cell width (successive-difference ratios 1.91 to 1.96, $p = 0.93$ to 0.97), as it did before the audit; a formally second-order scheme converging at first order in the EW points to a first-order error in the line-forming region whose source is not established. The continuum crossings are interpolated between the continuum points of 1.8 and 2.13. The 60 R_p domain factor measured before the audit (the EW lower by 1.7 per cent) was not measured again and is not applied.

How the grid is used (decision of 2026-09-24). The catalog keeps its 500-cell grid. A composition read from it is quoted as the continuum value of the table above, and the difference between the 500-cell and the continuum value is the grid uncertainty of every catalog number: from the EW, He/H = 1.84 (1.70 on the catalog grid, 0.033 dex lower, the 500-cell EW being about 6 per cent high); from the broadened line, about 1.93 (1.78 on the catalog grid).

The width of the line. The EW is matched, the width is not: the unbroadened red line of He/H = 1.8 is 4.18 per cent deep and 0.259 Å wide, against the fitted 1.24 per cent and the measured 0.841 Å. As in the p-winds comparison memo, the excess absorption of the instrument-convolved spectrum is convolved with a Gaussian of FWHM f in velocity, its depth, blue depth, ratio and red-pair FWHM are read with the three-Gaussian extractor used on the data, and f is matched where the FWHM reaches the measured 0.841 Å (the memo's atomic well-mixed He/H = 0.40 state is reproduced, $f = 22.22$ km s⁻¹; .P1/connectivity/broaden_molecular_final.py). MEASURED:

He/H	cells	EW, $f = 0$	matched f [km s ⁻¹]	red [%]	blue [%]	red/blue	EW at f
1.52	500	1.0149	22.19	1.133	0.148	7.64	0.9813
1.6	500	1.0577	22.18	1.181	0.155	7.63	1.0227
1.8	500	1.1550	22.15	1.289	0.170	7.60	1.1167
2.13	500	1.2902	22.12	1.440	0.190	7.57	1.2474
1.6	1000	1.0297	22.17	1.149	0.151	7.64	0.9955
1.8	1000	1.1249	22.14	1.255	0.165	7.61	1.0876
2.13	1000	1.2571	22.11	1.403	0.185	7.58	1.2155
1.6	2000	1.0150	22.16	1.133	0.148	7.64	0.9813
1.8	2000	1.1094	22.14	1.238	0.163	7.62	1.0726
2.13	2000	1.2403	22.10	1.384	0.182	7.59	1.1992
observed		1.108 ± 0.030		1.24 ^{+0.22} _{-0.23}	0.25 ^{+0.14} _{-0.12}	6.7 ^{+12.7} _{-3.1}	1.108 ± 0.030

Three readings of the table.

- *The kernel is set by the velocity field, not by the composition or the grid:* 22.10 to 22.19 km s⁻¹ on every state, the value the memo found for the atomic families and the one found here before the audit. At it the He/H = 1.8 line has depth, blue depth and ratio inside the observed 1 σ (Fig. 2).
- *The broadened EW moves the crossing up by about 0.08 in He/H:* the Gaussian wings carry absorption out of the red-pair window, lowering the EW by 3.3 per cent at the matched kernel.
- *As turbulence the kernel is supersonic.* On the advection-corrected profiles of He/H = 1.6 and 1.8, weighted over the line-forming region by $n(2^3S)r$, $T = 2650$ and 2756 K, $\mu = 1.94$ and 2.03 and $c_s = 4.33$ km s⁻¹; the matched $\sigma = 9.4$ km s⁻¹ is $\mathcal{M} = 2.2$ as a line-of-sight dispersion, 3.1 as $\sqrt{2}\sigma$ and 3.8 as an isotropic speed.

How the kernel is used (decision of 2026-09-24). The added Gaussian is an empirical parameter of the comparison, fixed by the measured width alone, and not a physical term of the model. Every composition this catalog reads from the full line is conditional on it: with the kernel, He/H = 1.78 on the 500-cell catalog grid and about 1.93 in the continuum limit of the grid; without it, from the EW alone, 1.70 and 1.84.

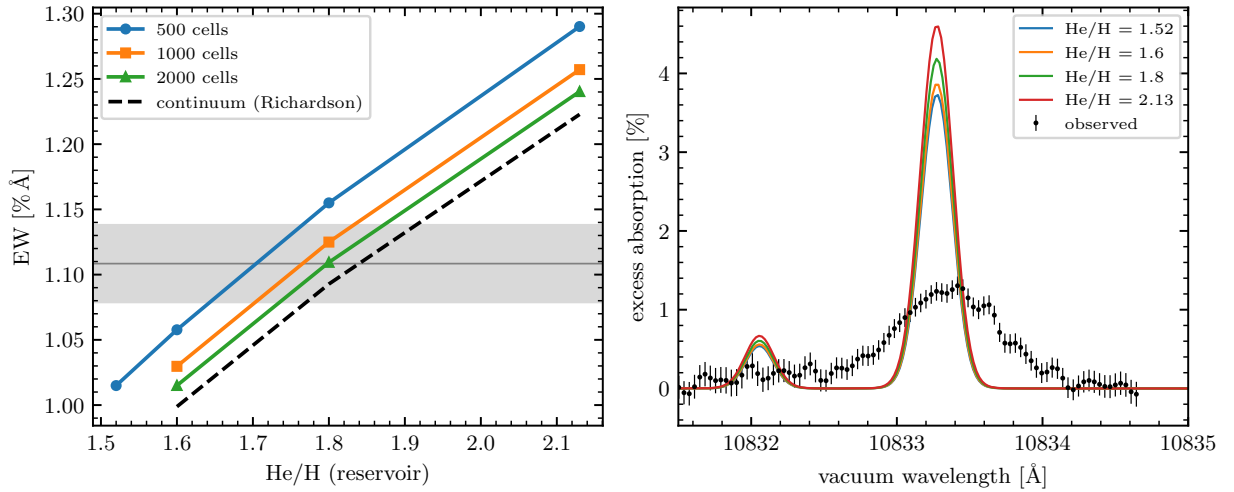


Figure 1: Left: the red-pair EW of the certified molecular states against the reservoir He/H on 500, 1000 and 2000 cells and in the continuum limit, with the observed 1.108 ± 0.030 % Å (grey). Right: the unbroadened 500-cell spectra against the released one.

6. What the flux-spread criterion misses, on one certified case

The question the columns above raise is what a state that meets $du < 10^{-3}$ but fails this code's certification differs from the certified stationary solution BY. Measured on the atomic scalar_kzz1e9/HeH2.13, which is certified, against a state of the same case stopped after one outer pass of the same route:

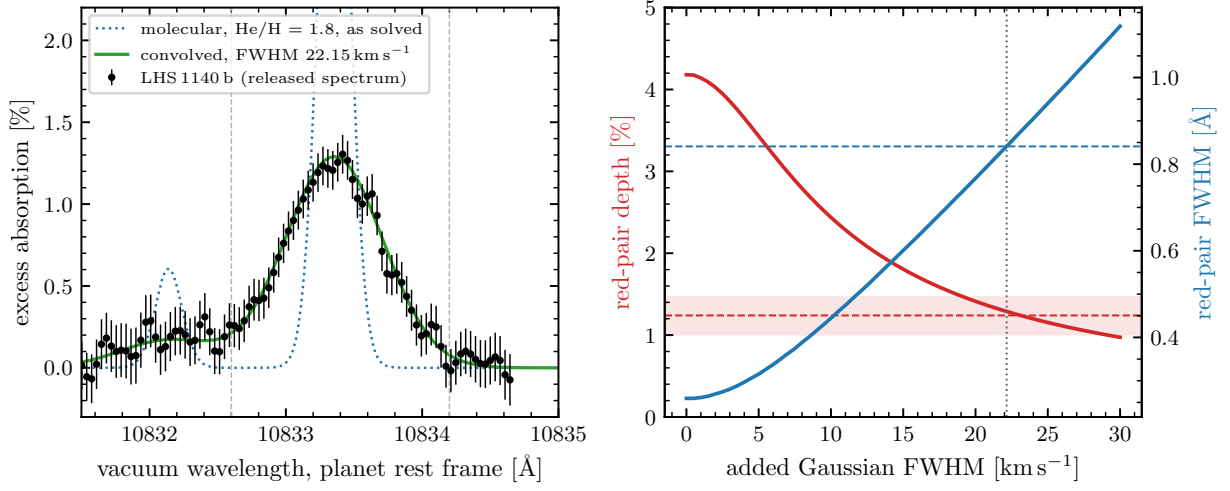


Figure 2: Left: the molecular $\text{He}/\text{H} = 1.8$ profile as solved (dotted) and convolved with a 22.15 km s^{-1} FWHM Gaussian (solid), against the released spectrum; both model curves are shifted by the memo's $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ for display (every EW is from the unshifted arrays), and the dashed verticals mark the EW window. Right: the red-pair depth (red, left axis) and FWHM (blue, right axis) as the added Gaussian is scanned; dashed lines are the observed values, the band the 1σ interval of the fitted depth, and the dotted vertical the matched kernel.

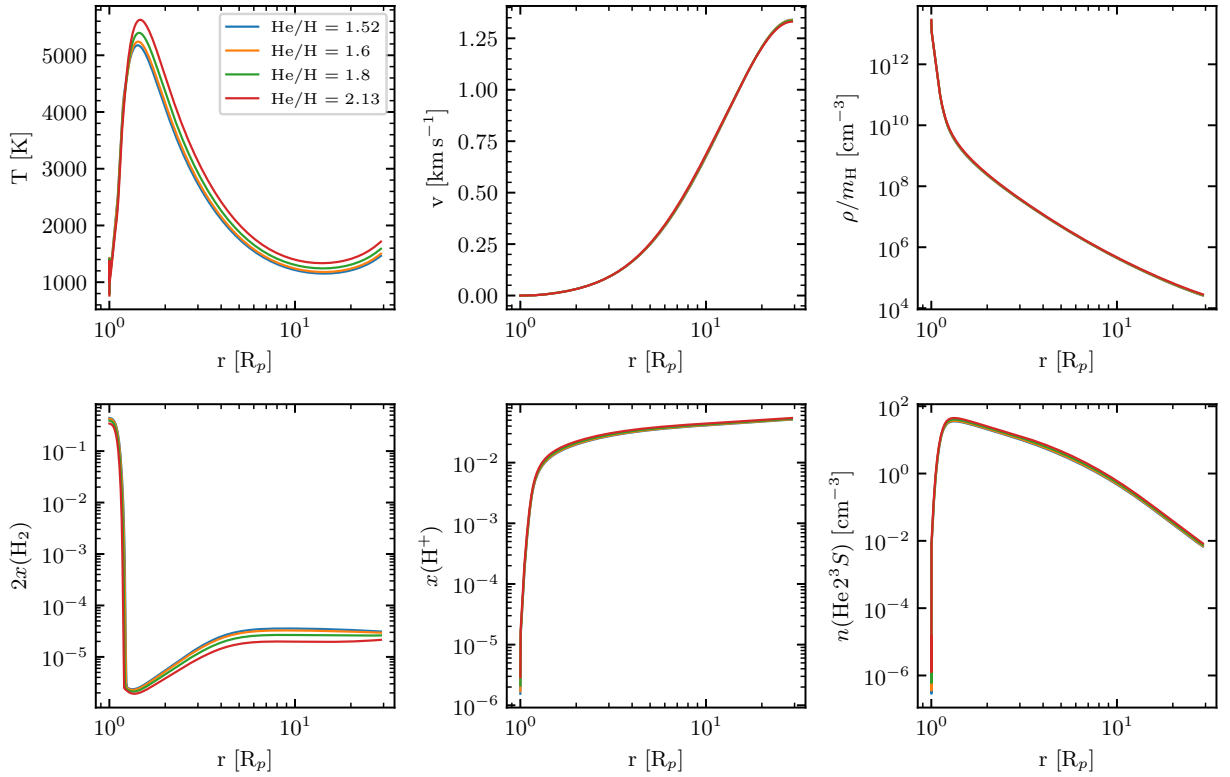


Figure 3: The certified molecular $K_{zz} = 10^9$ states $\text{He}/\text{H} = 1.52$ to 2.13 under the corrected atomic physics (advection-corrected profiles): temperature, velocity, ρ/m_{H} , H_2 fraction, H II fraction and $n(\text{He } 2^3\text{S})$ against radius.

	certified stationary	flux converged, refused
du , the run's own gate	2.46E-11	1.83E-12
$\ R\ $, re-evaluated	1.61E-08	4.97E-08
the three hydrodynamic rows	within	within
elemental transport He/H, $r \geq 1.2$	within	5.21E-05 above 1.0E-05 at $r = 1.695$
verdict	CERTIFIED	NOT CERTIFIED, one entry

Both clear the flux-spread criterion by four to eight decades and ONE row separates them. The two states differ by at most 1.9 per cent in any profile, and the difference sits in the composition and the temperature of the layer the He I 10830 line forms in (the metastable column's median radius is $6.4 R_p$, its 5 to 95 per cent range 1.7 to $16.7 R_p$), not in $\rho v r^2$, which agrees to 0.055 per cent everywhere. That is why the flux-spread criterion is blind to it.

It reaches a measured quantity. With the catalog's own selection the He I 10830 red-pair equivalent width moves from 1.374869 to 1.356745 % Å, a change of 1.32 per cent, and the red depth by 1.11 per cent, while $\log_{10} \dot{M}$ is 7.87 for both at the two decimals the code prints. For scale, the He/H rungs of this group stand 0.178 % Å apart in the equivalent width, so the shift is a tenth of the spacing the ladder measures a composition with.

Against the measurement it is smaller still. The observed red-pair equivalent width is 1.108 ± 0.030 % Å (Cherubim et al. 2026), and the two states differ by 0.018 % Å, which is 0.60 of one standard deviation. For a comparison with the data the two states are the same model: the refused one would be read as the same answer. What the certification buys is not a different observable but the statement that the state solves the elemental transport equation everywhere, which is what lets a later measurement, or a ladder that resolves 0.178 % Å in composition, rest on it.

LHS 1140 b, atomic_scalar_gj1132_kzz1e9/HeH2.13: two flux-converged states

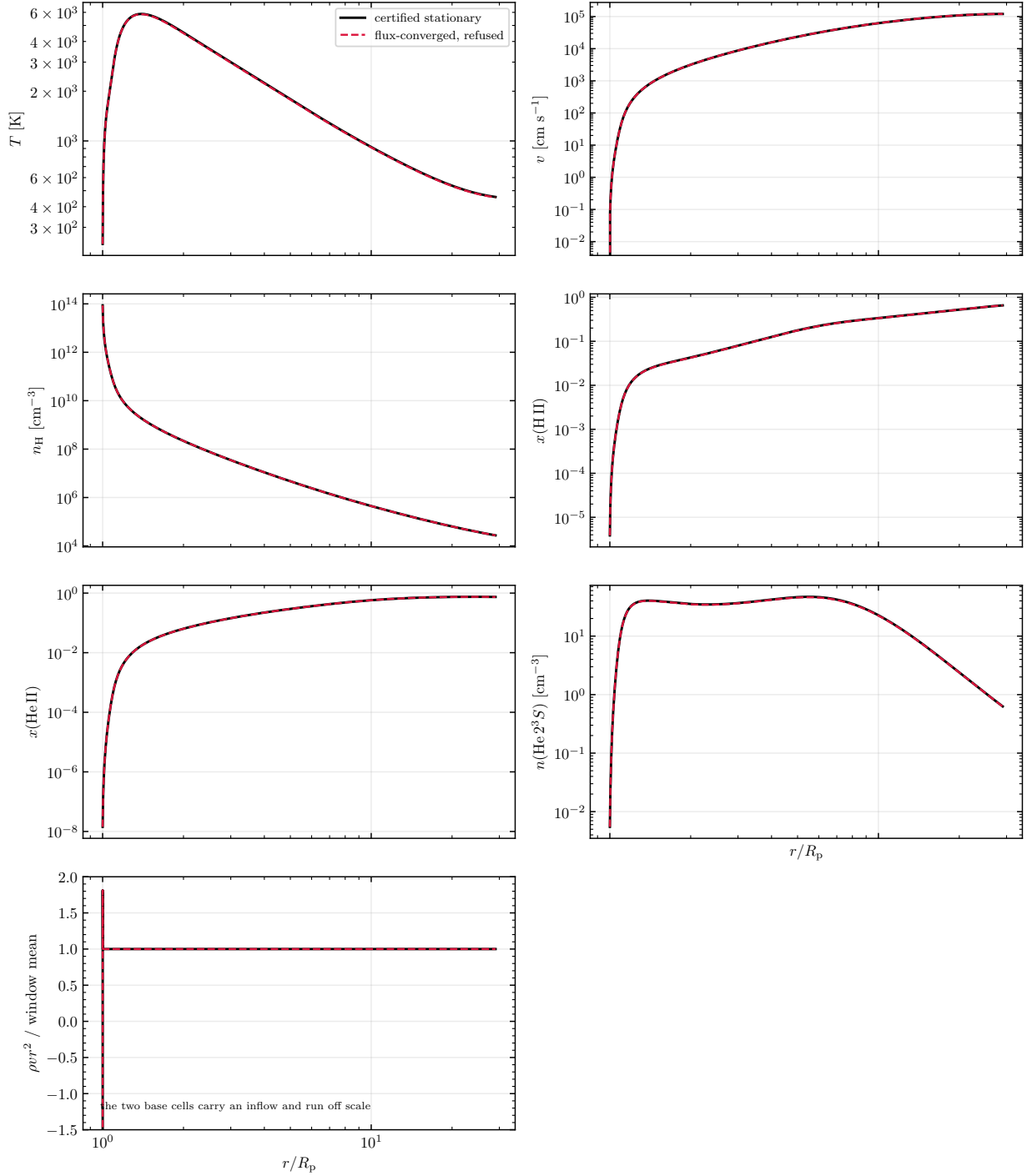


Figure 4: The certified stationary solution and the flux-converged refused state of the atomic scalar_gj1132_kzz1e9/HeH2.13, as functions of radius.

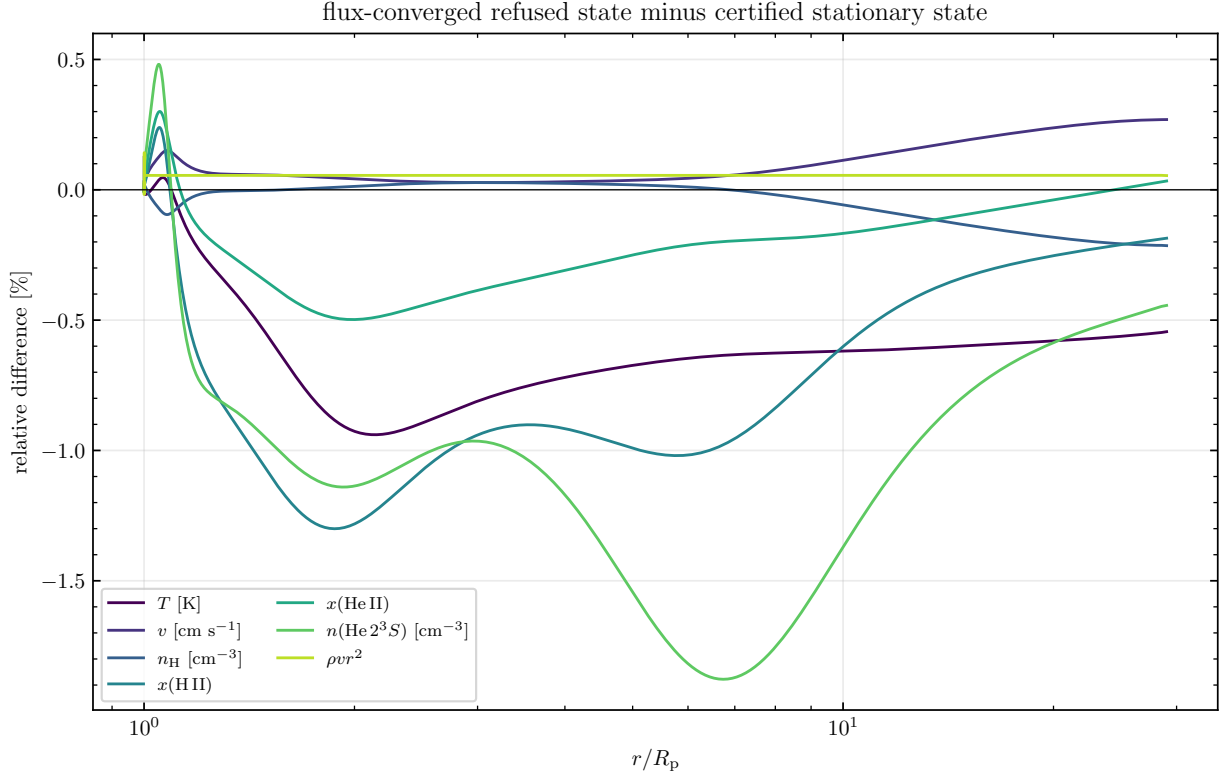


Figure 5: The same two states, as the relative difference of each quantity. The largest are the temperature at $r = 2.13 R_p$ (-0.94 per cent), $x(\text{H II})$ at 1.87 (-1.30) and $n(\text{He } 2^3S)$ at 6.74 (-1.88).

He I 10830 transmission, instrument convolved at $R = 68000$

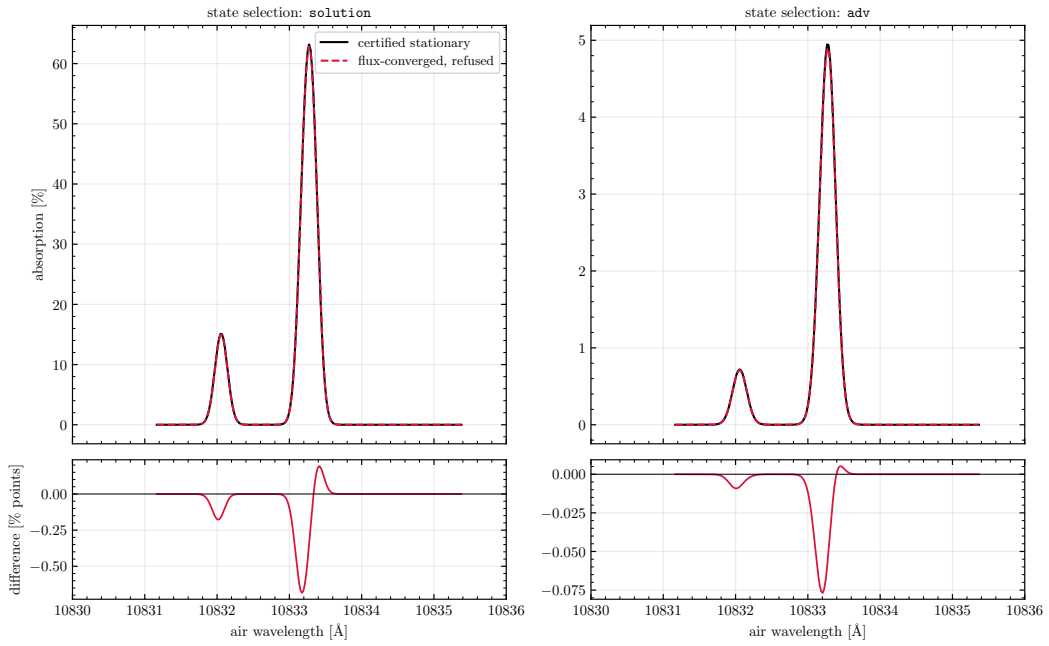


Figure 6: The He I 10830 transmission of the two states.